

DBMS LAB 0

Parth Janardan Naukudkar_ECS_II_66_D6

Aim: To study SQL commands, their uses, syntax, and install MySQL for database management experiments.

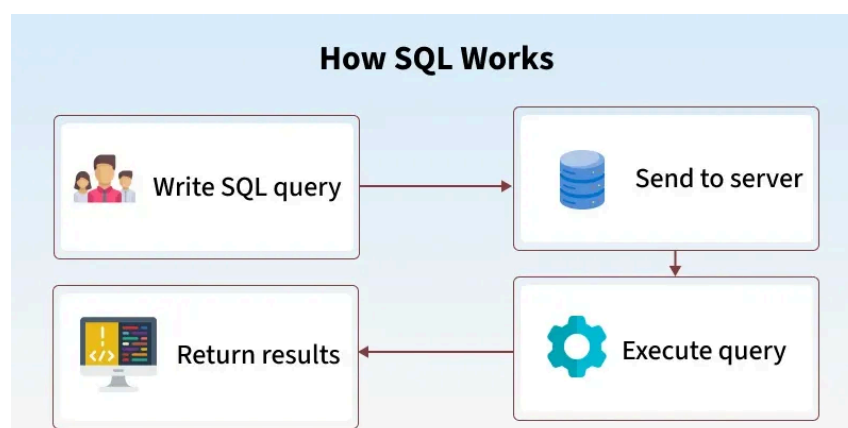
1. What is SQL and its Use?

SQL (Structured Query Language) is a standard programming language used to communicate with and manage databases. It helps users store, access, update, and organize data efficiently.

Uses of SQL

1. **Storing Data** – Add new information to a database.
2. **Retrieving Data** – Search and display required information.
3. **Updating Data** – Modify existing records.
4. **Deleting Data** – Remove unwanted records.
5. **Managing Databases** – Create and manage tables, records, and relationships.

Example: In a school database, SQL can be used to store student details and retrieve a student's **name, roll number, marks, and attendance**.



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2. Describe all the different commands of SQL, mention use of it and syntax.

Different SQL Commands, Their Uses and Syntax

SQL commands are instructions used to **create, manage, retrieve, and modify data** in a database. They are mainly divided into different categories based on their purpose.

1. DDL – Data Definition Language

DDL commands are used to **create and modify the structure of a database**.

Command	Use	Syntax
CREATE	Create a new database or table	CREATE TABLE table_name (column1 datatype, column2 datatype);
ALTER	Modifies the structure of an existing table.	ALTER TABLE table_name ADD column_name datatype;
DROP	Deletes a table or database permanently.	DROP TABLE table_name;
TRUNCATE	Removes all records from a table but keeps its structure.	TRUNCATE TABLE table_name;

Example:

CREATE TABLE Student (

Roll_No INT,

Name VARCHAR(50),

Marks INT

);

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2. DML – Data Manipulation Language

DML commands are used to **add, change, and remove data** from tables.

Command	Use	Syntax
INSERT	Adds new records to a table.	INSERT INTO table_name VALUES (...);
UPDATE	Modifies existing records.	UPDATE table_name SET column=value WHERE condition;
DELETE	Removes records from a table.	DELETE FROM table_name WHERE condition;

Example:

```
INSERT INTO Student VALUES (101, 'Rahul', 85);
```

3. DQL – Data Query Language

DQL is mainly used to retrieve data from a database.

Command	Use	Syntax
SELECT	Displays or retrieves data from a table.	SELECT column1, column2 FROM table_name;

Example:

```
SELECT Name, Marks FROM Student;
```

To retrieve all columns:

```
SELECT * FROM Student;
```

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4. DCL – Data Control Language

DCL commands are used to **control access and permissions** in a database.

Command	Use	Syntax
GRANT	Gives permissions to a user.	GRANT privilege ON table_name TO user;
REVOKE	Removes permissions from a user.	REVOKE privilege ON table_name FROM user;

Example:

```
GRANT SELECT ON Student TO user1;
```

5. TCL – Transaction Control Language

TCL commands are used to **manage transactions** in a database.

Command	Use	Syntax
COMMIT	Permanently saves the changes	COMMIT;
ROLLBACK	Undoes changes made in a transaction.	ROLLBACK;
SAVEPOINT	Creates a point to which changes can be rolled back.	SAVEPOINT savepoint_name;

Example:

```
UPDATE Student SET Marks = 90 WHERE Roll_No = 101;
```

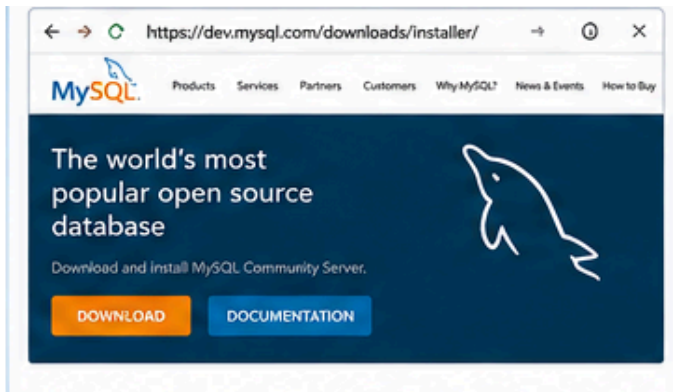
```
COMMIT;
```


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3. Write steps by steps procedure to install MYSQL.

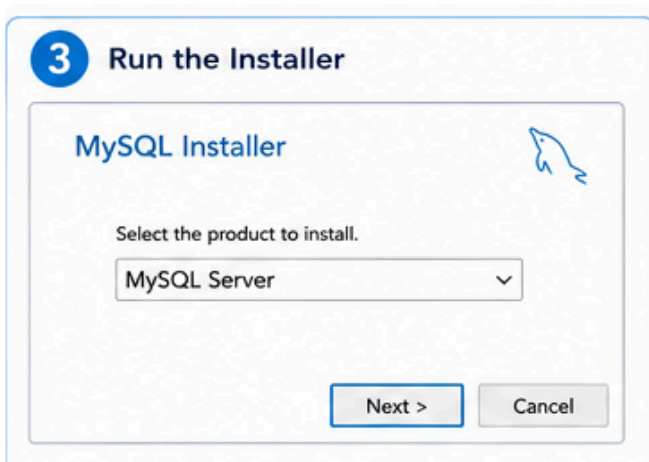
Step 1: Go to MySQL Official Website



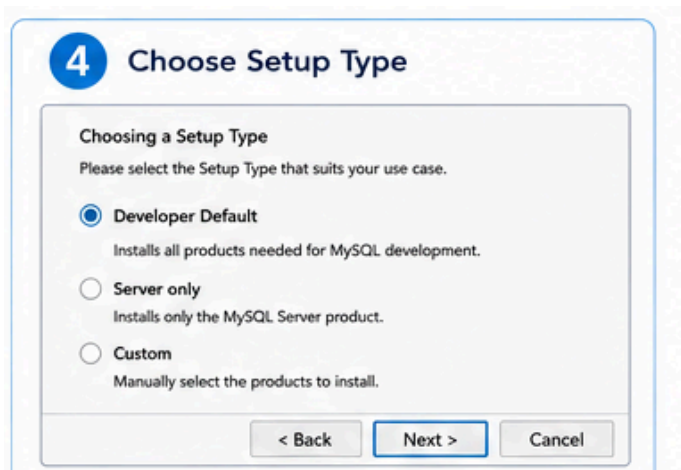
Step 2: Download MySQL Installer



Step 3: Run the installer



Step 4: Choose Setup Type



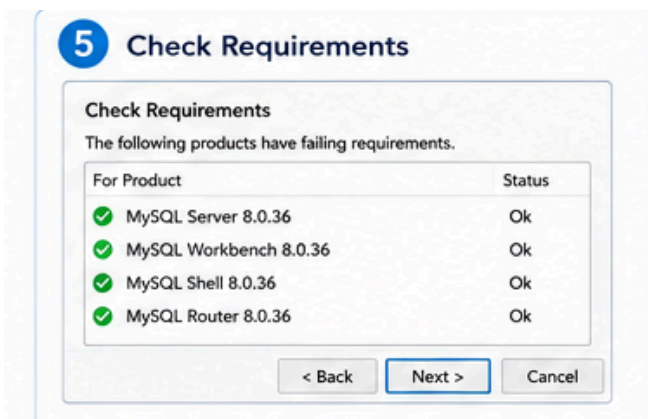
4 Choose Setup Type

Choosing a Setup Type
Please select the Setup Type that suits your use case.

- ☒ **Developer Default**
Installs all products needed for MySQL development.
- ☐ **Server only**
Installs only the MySQL Server product.
- ☐ **Custom**
Manually select the products to install.

< Back **Next >** Cancel

Step 5: Check Requirements



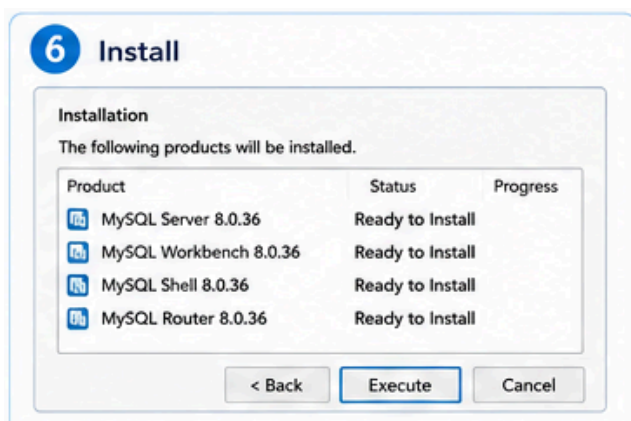
5 Check Requirements

Check Requirements
The following products have failing requirements.

For Product	Status
✓ MySQL Server 8.0.36	Ok
✓ MySQL Workbench 8.0.36	Ok
✓ MySQL Shell 8.0.36	Ok
✓ MySQL Router 8.0.36	Ok

< Back **Next >** Cancel

Step 6: Install



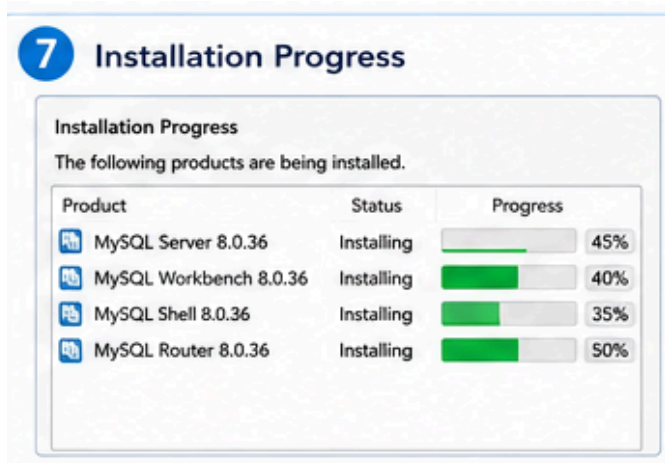
6 Install

Installation
The following products will be installed.

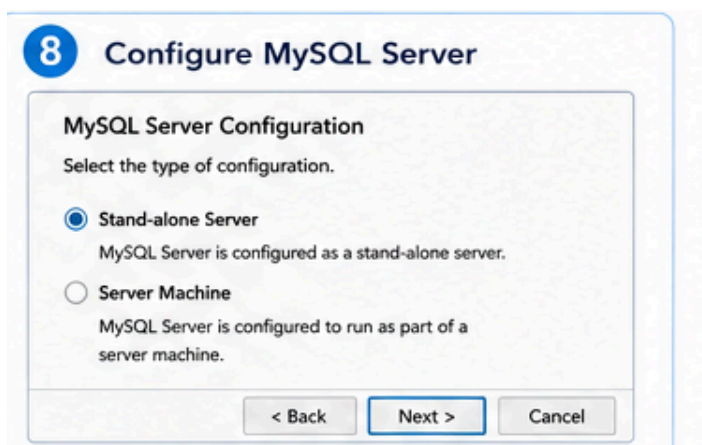
Product	Status	Progress
MySQL Server 8.0.36	Ready to Install	
MySQL Workbench 8.0.36	Ready to Install	
MySQL Shell 8.0.36	Ready to Install	
MySQL Router 8.0.36	Ready to Install	

< Back **Execute** Cancel

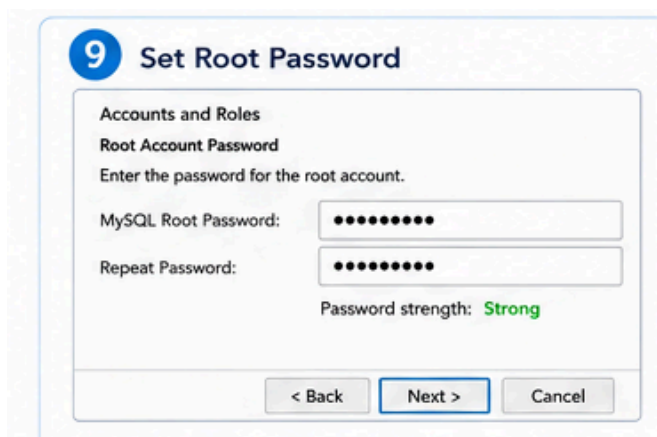
Step 7: Installation Process



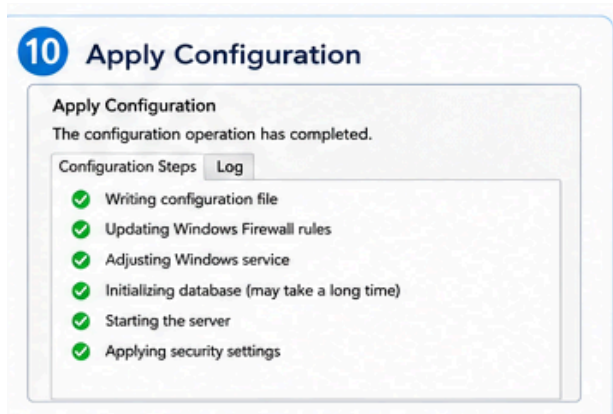
Step 8: Configure MySQL Server



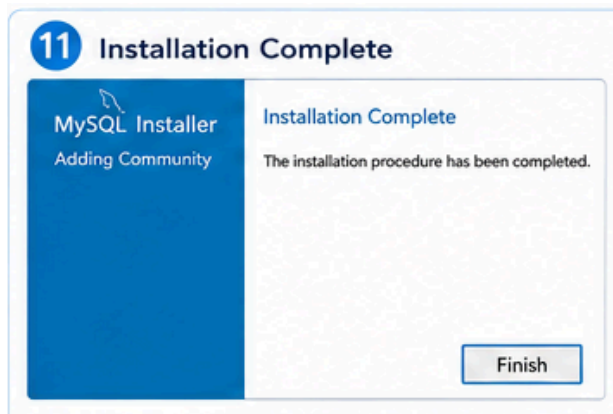
Step 9: Set Root Password



Step 10: Apply Configuration



Step 11: Installation Complete



Step 12: Verify MySQL Installation

